SafeHome AI

Complete Product, UX, AI, Data, and Technical Design Specification

Purpose

A self-contained implementation specification detailed enough for a developer or a new AI coding assistant to understand, reproduce, test, and extend the application without access to the original development conversation.

Product mission

Help older adults, families, and caregivers turn one room photograph into understandable evidence, focused human follow-up, a bounded environmental risk score, and practical actions - while preserving human judgment and protecting room-photo privacy.

Status: implementation-aligned specification

SafeHome AI is an educational environmental-safety tool. It does not diagnose a medical condition, predict an individual person's probability of falling, certify a room as safe, or guarantee fall prevention.

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1. Product definition and principles

Product summary

SafeHome AI reviews one room photo for possible environmental fall hazards. It checks photo quality, asks vision AI for visible evidence and approximate coordinates, asks only the follow-up questions the image cannot answer, calculates a 0-100 room score, and generates a prioritized safety plan. Saving, history, comparisons, reports, and email sharing are optional.

Core principles

Senior-friendly without removing functionality; quick use before account creation; AI proposes while application code validates; a person confirms uncertainty; photos remain session-only; lower scores represent fewer identified concerns; results are educational and require human review.

Language rules

Use “Room Name,” never “Room ID,” in the UI. Use “account” or “email address,” never “Home ID.” Use “Follow-Up Questions,” never “checklist.” Avoid technical language, repeated explanations, redundant popups, and nested score expanders.

2. Users and end-to-end journeys

Guest / Quick Safety Check

Choose room type -> upload one photo -> review quality -> analyze -> review marked hazards -> answer only relevant follow-up questions -> see score and three actions -> optionally compare after fixes, download, email, or choose Save This Check. No account or Room Name is required until saving.

Account / Full Safety Check

Choose room type -> sign in or create account -> select or create a Room Name -> upload and analyze -> answer follow-ups -> view score and plan -> save automatically once -> view room history, trend, before/after categories, and email summaries.

Recheck journey

From room statistics, select Analyze This Room Again. The app restores the room type and Room Name, accepts a new image, completes the full analysis, compares it to the latest saved check, and stores the completed recheck exactly once.

3. Navigation and screen specifications

Router

Streamlit session-state routes: landing, room_selection, room_id_selection, photo_upload, ai_results, checklist (internal name only), checklist_summary, risk_score, safety_report, after_fixes_photo, saved_results, and room_stats. All non-landing pages expose one global Back to Landing Page action.

Landing page

Show SafeHome AI, the tagline “Find home fall hazards before they become accidents,” a four-step explanation, optional signed-in status and logout, Privacy and AI information, How SafeHome AI works, Quick Safety Check, Full Safety Check, My Saved Room Checks when signed in, and Room-by-Room Stats. Do not put Continue Without Signing In on the landing page.

Room selection

Offer Living Room, Bedroom, Bathroom, Kitchen, Hallway, Stairs, Dining Room, Laundry Room, Garage, Entryway / Foyer, Basement, Outdoor Walkway / Porch, Closet / Storage Area, Home Office, and Other. Quick mode goes directly to upload; full mode goes to Room Name selection.

Progress presentation

Use Step 1 Choose a Room; Step 2 Upload Room Photo; Step 3 AI Results; Step 4 AI Follow-Up Questions; Step 5 Your Safety Plan; Step 6 Your Safety Plan & Report. Each screen has one clear primary action and secondary actions below it.

4. Accounts, authentication, and Room Names

Identity

The account identifier is a valid email address, normalized to lowercase. The Existing Account tab appears before Create New Account. Account creation combines availability checking with Create Account; there is no separate availability button.

Passwords

Minimum eight characters. Generate a random hexadecimal salt. Store PBKDF2-HMAC-SHA256 with 300,000 iterations as salt$digest. Compare hashes in constant time. Never store or log readable passwords.

Recovery

The user enters the account email, selects Email Reset Code, receives a six-digit code at the stored recovery email, and has 15 minutes to enter it with a new password. Store only SHA-256 of the code. A new request replaces the old code; only the newest code works. Clear reset fields after success.

Room Names

User-defined, preserve capitalization, collapse repeated whitespace, length 1-80 characters. Examples: Bedroom 1, Guest Bedroom, Main Bathroom. If no rooms of the selected type exist, open Create New Room Name directly. Otherwise show Use Existing Room Name first, then Create New Room Name. Unsigned users may continue without signing in on this page.

5. Photo intake and local quality review

Upload contract

Exactly one JPG, JPEG, PNG, or WEBP image; 30 MB maximum; iPhone-friendly photo library / camera behavior. Correct EXIF orientation with Pillow, preserve aspect ratio, show an upright full-width preview, and keep photo bytes only in session state.

Local metrics

Measure width, height, grayscale mean brightness, grayscale standard-deviation contrast, and edge standard deviation as a sharpness estimate. These heuristics do not identify people or objects.

Thresholds

Very low resolution: width below 400 or height below 300. Slightly low: below 640 or 480. Very dark: brightness below 45; dark: below 70. Washed out: above 235; bright: above 215. Very low contrast: below 15; low: below 25. Very blurry: edge score below 5; slightly blurry: below 9.

Quality label

Poor for two or more severe conditions; Caution for one severe condition or at least two total issues; Good otherwise. A poor result warns and suggests a retake but does not necessarily block analysis.

6. Vision AI contract and guardrails

Modes

Real mode requires AI_ANALYSIS_MODE=real, OPENAI_API_KEY, and OPENAI_MODEL. Sample mode intentionally uses built-in results. If real analysis fails, fallback mode returns built-in results clearly labeled as samples, never as findings from the uploaded photo.

Inspector mindset

Mentally walk likely paths from the camera/doorway toward visible furniture, bed, chair, stairs, bathroom, or kitchen. Consider foot catches, sliding, twisting, blocked routes, sharp turns, reaching, bending, support stability, visibility from standing height, and nearby consequences of a stumble.

AI restrictions

Do not identify people; infer age, disability, diagnosis, or health; mention names, addresses, mail, labels, prescriptions, or private documents; invent unsupported hazards; say a room is safe; or guarantee prevention. Prefer three to six visible or strongly supported hazards. Move important uncertainty into questions.

Result fields

Return summary; hazards containing category, title, confidence, visibility, risk_points, location, location_x, location_y, evidence, explanation, recommendation, and human_review_reason; not_visible; checklist_questions; skip_buffer_points; and safety_reminder. JSON only.

Application validation

Maximum eight rendered hazards, six questions, and five not-visible items. Unknown categories become unclear. Invalid confidence becomes medium. Coordinates require integer pairs from 0 through 100. Hazard points require 1 through 15; otherwise use the category backup. Clamp skip buffer from 0 through 15 and force it to zero when there are no questions.

Allowed hazard taxonomy and category backup values

Internal category	Visible label	Backup points
loose_rug	Loose Rug or Mat	10
cords	Cord Hazard	12
clutter	Floor Clutter	10
poor_lighting	Poor Lighting	8
slippery_floor	Slippery or Wet Floor	12
narrow_pathway	Narrow or Blocked Pathway	8
stairs	Stairs or Step Hazard	15
handrail	Handrail Concern	15
bathroom_grab_bars	Bathroom Grab Bar Concern	15
hard_to_reach_items	Hard-to-Reach Items	6
threshold_trip	Raised Threshold / Floor Transition	10
unstable_furniture	Unstable Furniture	10
pet_items	Pet Items in Walking Path	6
footwear	Footwear in Walking Path	6
low_seating	Low or Unsafe Seating	8
poor_contrast	Poor Contrast / Hard-to-See Edges	6
uneven_floor	Uneven Floor Surface	12
door_mat	Loose Door Mat	8
furniture_in_path	Furniture Blocking Walking Path	8
outdoor_surface	Outdoor Surface Hazard	12
laundry_on_floor	Laundry on Floor	8
open_drawers_cabinets	Open Drawers or Cabinets	8

7. Hazard display and visual markers

Coordinates

AI returns x and y percentages where 0,0 is top-left and 100,100 is bottom-right. Draw a blue outline circle with its hazard number, no red, no white center dot, and no marker when coordinates are invalid. Preserve image proportions and explain that the marker is approximate.

Hazard card

Use native Streamlit controls to prevent raw HTML. Show Hazard N and category, title, priority, Why it matters, What the app noticed, Suggested fix, optional location/confidence/visibility caption, and a real-room double-check note. Do not show per-card model scoring or “AI severity assessment.”

Priority

Fix Now: cords, slippery floors, stairs, handrails, grab bars, thresholds, uneven floors, outdoor surfaces, and open drawers. Fix Soon: rugs, clutter, narrow paths, lighting, reaching, unstable furniture, mats, blocked paths, laundry, low seating, and poor contrast. Watch/Review: pet items, footwear, and unclear findings. Do not duplicate these in a separate urgent section.

8. Adaptive follow-up questions

Generation

Questions are created from important uncertainties in the specific image, not from a premade universal form. Ask zero to six questions in short, simple English. Every question is a positive safety condition so No means the concern exists.

Answers

Exactly Yes, No, Not sure, and Not applicable. Standardize the lighting question as “Does the room have adequate lighting at night?” Show question number, total, progress, reason, optional Read Aloud, Save and Next, Skip This Question, Skip Follow-Up Questions, and Previous Question after the first.

Skipping

Skipping one question records it as skipped/not applicable. Skipping all questions clears answers and applies the AI-generated uncertainty buffer. The UI must say Skip Follow-Up Questions, never Skip Entire Checklist.

9. Scoring, risk levels, and action ranking

Hazard points

AI assigns each separate visible hazard 1-15: 1-3 minor/off-path, 4-6 low concern, 7-9 clear walking-path concern, 10-12 serious likely trip/slip concern, and 13-15 urgent/high-consequence environmental concern. The same photo and condition should remain within a narrow scoring band. Invalid values use the category table below.

Follow-up points

For a positive-condition question, Yes adds 0; No adds 80% of the category backup; Not sure adds 35% of that 80%; Not applicable adds 0. If every question is skipped, add the AI buffer once, clamped 0-15.

Formula

Raw score = sum of valid AI hazard points + follow-up concern points + optional skip buffer. Final score = rounded raw score capped at 100. Risk labels: 0-29 Low Risk, 30-59 Moderate Risk, 60-100 High Risk. Higher means more possible environmental hazards; it is not a medical probability.

Presentation

Show numeric score, written risk label, and a green/amber/red severity bar with Low, Moderate, and High text. Use one expander titled How this score was calculated. It contains AI points, AI-assessed count, category-backup count, follow-up points, skip buffer, raw score, final score, and risk label. Do not nest another breakdown expander.

Action plan

Rank Fix Now before Fix Soon before Watch/Review; overlap between AI and follow-up before non-overlap; higher category points next; stable text order last. Remove duplicate recommendations. Show three actions immediately and optional extra steps. Estimate improvement as roughly 50%-100% of category points. Recommend help for handrails, grab bars, stairs, and uneven-floor repairs.

10. Reports, email, and before/after comparison

Final report

Include room type, optional Room Name, AI summary, hazards, follow-up answers, score and label, recommendations, educational limitation, and human-review reminder. Allow Read Aloud, text download, email, copy/export, and Start New Safety Check.

Email recipients

Apply the same component everywhere. Always keep one field. Default the first to the signed-in email. Add new email address up to five total. Put an aligned X beside each field, including on mobile. Validate all addresses and reject duplicates. Send text only; never attach the photo.

Email destinations

Support final safety report, room trend, saved before/after comparison, and room-statistics summary. Prefer server-side Brevo delivery. If disabled or failed, keep a mailto draft and copyable body. Put technical errors in an optional detail area.

Session comparison

Compare After Making Fixes retains the original photo, hazards, and score in session. Analyze a second image and show before score, after score, score change, resolved categories, still needing attention, new categories, and new-image markers. Use This New Check as Current Results replaces the active analysis, resets follow-up progress, increments the check nonce, and can be saved as a new history entry.

11. Saved history, statistics, and trends

Saving

Every completed signed-in check with a Room Name is automatically saved exactly once. A per-run nonce prevents Streamlit reruns from creating duplicates. This includes rechecks and accepted after-fixes checks. Quick checks stay session-only until Save This Check sends the user through account and Room Name selection.

Saved checks

Show total checks, average score, most common room, and individual cards with sequential Check N, room type, Room Name, score, risk label, and readable date. Never expose UUIDs or raw timestamps.

Dashboard

Rooms Checked counts distinct Room Names with at least one check, not number of checks. Show most common hazard without a count, latest improvement, next room to check, and total saved checks as secondary information. Do not show Rooms Needing Attention.

Room table

Columns: Room Name, Room Type, Checks, Average Score, Latest Risk, Top Hazard. Selected-room details: check count, average, latest, highest, lowest, and latest risk label. Check history columns: Checked, Score, Risk Label, Hazards. Format times like Jul 30, 2026 at 2:16 PM UTC.

Trend

Require two checks. Show first, latest, direction, and plain-language message. Because higher is worse, a decrease is improvement and an increase needs attention. Use a theme-aware SVG chart with 0/50/100 grid lines, Check N labels, and score/100 values beside every point.

Saved comparison

Require two checks for one Room Name. Dropdown labels contain readable date, score, and risk label, never internal IDs. Show before/after score and metadata plus Resolved, Still Present, and New groups, with download, copy, and email options.

12. Accessibility and visual design system

Global panel

Collapsed by default and labeled Accessibility. Settings apply immediately: Standard/Large/Extra Large text; System/Light/Dark/High Contrast color; Show Read Aloud buttons; Reduce motion and animations; Use compact spacing.

Read Aloud

Use browser speechSynthesis for AI results, individual follow-up questions, top fixes, and final report. Cancel existing speech before starting new speech.

Themes

Style backgrounds, text, buttons, disabled states, inputs, placeholder text, dropdowns and their option menus, tabs, expanders, help icons, tooltips, tables, charts, and upload controls. Dark and High Contrast must never become white-on-white or black-on-black. High Contrast uses black, white, yellow, and cyan with strong borders.

Responsive behavior

Use single-column flow on iPhone where practical, minimum 44-pixel touch targets, no clipped text, no viewport overflow, horizontally aligned email field/X pairs, stack metrics gracefully, and convert tables to readable responsive rows. Compact spacing may reduce whitespace but not touch-target size.

Brand tone

Calm, practical, trustworthy, non-clinical, non-alarmist, and not childish. Primary blue with navy/slate structure and mint accents. Risk colors always include text: green Low, amber Moderate, red High.

13. Data model, privacy, and security

Photo policy

Photos are used for preview, quality review, AI analysis, marker rendering, and active-session comparison. They are not stored in Supabase and are not attached to email. Warn users to omit faces, addresses, mail, bills, medication containers, and medical records.

Conceptual tables

homes: account email, password hash, recovery email, reset hash and expiry, created time. home_rooms: UUID, account email, Room Name, room type, display label, created time. room_checks: check UUID, account, room fields, score, label, hazard count, answer counts, skipped flag, analysis mode, app version, created time. room_check_details: typed AI hazard, follow-up, and recommended-fix records.

Security

Keep secrets in environment variables or Streamlit secrets. Validate emails, Room Names, AI categories, coordinates, and points server-side. Escape AI text before custom HTML and prefer native components. Never log passwords, hashes, reset codes, photo data, or API keys. Require authentication for saved records.

14. Architecture, configuration, and source layout

Services

Streamlit: navigation and accessible UI. Pillow: orientation, validation, quality metrics, and marker drawing. OpenAI: image evidence, coordinates, questions, hazard severity, and uncertainty buffer. Supabase: accounts, rooms, history, details, comparisons. Brevo: reset and summary emails. Browser speech synthesis: optional voice output.

Configuration

AI_ANALYSIS_MODE, OPENAI_API_KEY, OPENAI_MODEL, DATABASE_ENABLED, SUPABASE_URL, SUPABASE_KEY, EMAIL_ENABLED, BREVO_API_KEY, EMAIL_SENDER_ADDRESS, EMAIL_SENDER_NAME, and APP_VERSION. Guest/sample use must remain functional when optional services are absent.

Modules

app.py is only the entry point. router.py owns route setup. page_handlers.py owns workflows. app_state.py owns session resets. constants.py owns taxonomy. ui.py owns styles and reusable presentation. account_ui.py handles auth. database.py handles Supabase. saved_checks.py handles save orchestration. image_tools.py and photo_quality.py handle images. ai_analysis.py normalizes AI. scoring.py calculates scores. priorities.py and fixes.py rank actions. report_builder.py, email_service.py, email_ui.py, trends.py, and comparison.py own their named concerns.

Session state

Track page, room_type, photo and quality state, ai_result, question index and answers, skipped flag, score and breakdown, report, accessibility preferences, save state and IDs, signed-in account, current Room Name, upload nonce, photo bytes, quick mode, before/after state, check_run_nonce, and database_saved_run_nonce. Reset check state without clearing account or accessibility preferences.

15. Edge cases and definition of done

Graceful failure

Invalid/large/unreadable image produces a clear correction. No hazards does not imply safety. No questions allows direct scoring. Database outage preserves guest analysis. Email outage preserves download and mailto backup. Save failure preserves current work. AI outage produces labeled sample fallback. Raw provider errors belong in Technical details.

Streamlit correctness

Every repeated widget has a unique key. Reruns do not duplicate saves, reset account state, duplicate sign-in status, lose the photo, repeat success messages, or recreate recipient fields. Avoid deprecated use_container_width and components HTML APIs.

Completion criteria

Both Quick and Full flows work; authentication and reset work; one 30 MB image uploads on iPhone; orientation and quality checks work; AI output is validated; blue markers align; adaptive questions work; scoring remains 0-100 with 1-15 hazards; one score expander exists; three actions rank correctly; every signed-in check saves once; statistics count rooms correctly; dates are readable; charts are themed; emails accept 1-5 recipients; comparisons work; photos remain unsaved; Read Aloud works; four themes are readable; mobile layouts are unclipped; no raw HTML, UUIDs, duplicate widgets, obsolete warnings, or unnecessary popups remain.